Simplifying algebraic expressions

Collecting like terms, and taking a common factor out of a bracket.

LESSON 11–12 | 2 × 40 MIN | MATEMATIKA 6, §2 | S7 2.3

LESSON CLOCK

20' | 25' | 20' | 12' | 3

Like terms	20 min
Collecting	25 min
Expanding a bracket	20 min
Taking a factor out	12 min
Homework	3 min

LEARNING OBJECTIVES

- Identify like terms and collect them.
- Expand a bracket by multiplying every term inside it.
- Take a common factor outside a bracket.
- Check a simplification by substituting a number.

TEXTBOOK

National	pp. 29–34
Cambridge	Stage 7, pp. 25–29
Workbook	Exercise 2.3

01 Explanation

Like terms

Terms are **like** when their letter parts are identical. Only like terms can be collected — for the same reason that apples and oranges cannot be added into one number of apples.

PAIR	LIKE?	WHY
$5x$ and $-2x$	yes	the same letter
$5x$ and $5y$	no	different letters
$5x$ and $5x^2$	no	different powers
$3ab$ and $7ba$	yes	$ab = ba$

THE COEFFICIENT MAY DIFFER; THE LETTERS MAY NOT

$5x$ and $-2x$ are like terms and collect to $3x$. $5x$ and $5y$ are not, and $5x + 5y$ is already as simple as it gets.

Collecting

EXPRESSION	COLLECTED
$5x + 3x$	$8x$
$7a - 2a$	$5a$
$4x + 3y + 2x - y$	$6x + 2y$
$6m - 9m$	$-3m$
$3a + 4b$	$3a + 4b$

A TERM CARRIES THE SIGN IN FRONT OF IT

In $4x + 3y + 2x - y$ the last term is $-y$, with coefficient -1. Collecting the y terms gives $3y - y = 2y$, not $4y$.

Expanding a bracket

$$a(b + c) = ab + ac$$

EXPRESSION	EXPANDED
$3(x + 4)$	$3x + 12$
$5(2a - 3)$	$10a - 15$
$x(x + 2)$	$x^2 + 2x$
$2(x + 3) + 4x$	$6x + 6$

EVERY TERM INSIDE THE BRACKET IS MULTIPLIED

$3(x + 4)$ is $3x + 12$, not $3x + 4$. Drawing two arrows from the 3 to both terms stops this error for good.

Taking a factor out

Expanding read backwards: find what every term has in common and write it outside.

EXPRESSION	COMMON FACTOR	FACTORISED
$6x + 9$	3	$3(2x + 3)$
$4a - 10$	2	$2(2a - 5)$
$x^2 + 5x$	x	$x(x + 5)$
$12m + 8n$	4	$4(3m + 2n)$

CHECK BY SUBSTITUTING A NUMBER

Put $x = 2$ into $6x + 9$ and into $3(2x + 3)$: both give 21. One substitution catches almost every slip in this lesson.

02 Worked examples

EXAMPLE 1 **Simplify $4x + 3y + 2x - y$.**

1. x terms: $4x + 2x = 6x$.
2. y terms: $3y - y = 2y$.
   The last term is $-y$.
3. $= 6x + 2y$
   Nothing else is like.

ANSWER

$6x + 2y$

EXAMPLE 2 **Simplify $2(x + 3) + 4x$.**

1. Expand: $2x + 6 + 4x$.
   Both terms multiplied.
2. Collect: $6x + 6$.
3. Check at $x = 1$: $8 = 8$ ✓

ANSWER

$6x + 6$

EXAMPLE 3 **Factorise $6x + 9$.**

1. Both terms divide by 3.
2. $6x \div 3 = 2x$ and $9 \div 3 = 3$.
3. $= 3(2x + 3)$
   Check by expanding.

ANSWER

$3(2x + 3)$

03 Interactive model

Use coloured chalk for each letter when collecting; the class sees the groups before it counts them, and the sign errors halve.

Collect, expand, factorise

$5x + 3x$ equals:

- $8x$
- $8x^2$
- $15x$
- 8

$5x$ and $5y$ collect to:

- $10xy$
- $10x$
- they do not collect
- $25xy$

$4x + 3y + 2x - y$ equals:

- $6x + 4y$
- $6x + 2y$
- $8x + 2y$
- $6x - 2y$

$6m - 9m$ equals:

- $3m$
- $-3m$
- $15m$
- $-15m$

$3(x + 4)$ equals:

- $3x + 4$
- $3x + 12$
- $x + 12$
- $3x + 7$

$5(2a - 3)$ equals:

- $10a - 3$
- $10a - 15$
- $7a - 15$
- $2a - 15$

$6x + 9$ factorises to:

- $3(2x + 3)$
- $3(2x + 9)$
- $6(x + 9)$
- $3(x + 3)$

$x^2 + 5x$ factorises to:

- $x(x + 5)$
- $x(x + 5x)$
- $5x(x + 1)$
- $x^2(1 + 5)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To simplify	Soddalashtirish	Упростить
Like terms	O‘xshash hadlar	Подобные члены
To collect	Ixchamlash	Приведение подобных
To expand	Qavsni ochish	Раскрыть скобки
Common factor	Umumiy ko‘paytuvchi	Общий множитель
Bracket	Qavs	Скобка
Coefficient	Koeffitsiyent	Коэффициент
Substitution check	Tekshirish	Проверка подстановкой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Like terms have:

- the same coefficient
- the same letter part
- the same sign
- nothing in common

$7a - 2a$ equals:

- $5a$
- $9a$
- 5
- $14a$

$3a + 4b$ simplifies to:

- $7ab$
- $7a$
- it is already simplest
- $12ab$

$x(x + 2)$ equals:

- $x^2 + 2$
- $x^2 + 2x$
- $2x^2$
- $x + 2x$

$4a - 10$ factorises to:

- $2(2a - 5)$
- $2(2a - 10)$
- $4(a - 10)$
- $2(a - 5)$

A simplification is checked by:

- looking
- substituting a number
- expanding again
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $5x + 3x$

ANSWER $8x$

2. $7a - 2a$

ANSWER $5a$

3. $6m - 9m$

ANSWER $-3m$

4. $3(x + 4)$

ANSWER $3x + 12$

5. $5(2a - 3)$

ANSWER $10a - 15$

6. $6x + 9$ factorised

ANSWER $3(2x + 3)$

7. $3a + 4b$

ANSWER Already simplest

Medium · the standard question

7 PROBLEMS

1. $4x + 3y + 2x - y$

ANSWER $6x + 2y$

2. $2(x + 3) + 4x$

ANSWER $6x + 6$

3. $x(x + 2)$

ANSWER $x^2 + 2x$

4. $x^2 + 5x$ factorised

ANSWER $x(x + 5)$

5. $12m + 8n$ factorised

ANSWER $4(3m + 2n)$

6. $5a + 2b - 3a + 4b$

ANSWER $2a + 6b$

7. $3(2x - 1) + 5$

ANSWER $6x + 2$

Hard · stretch and reason

7 PROBLEMS

1. $4(x + 2) - 3(x - 1)$

   ANSWER $x + 11$

2. $2x(x + 3) - x^2$

   ANSWER $x^2 + 6x$

3. $5(a + b) - 5a$

   ANSWER $5b$

4. $3x + 2y - (x - y)$

   ANSWER $2x + 3y$

5. Factorise $9x^2 + 6x$

   ANSWER $3x(3x + 2)$

6. Check $6x + 9 = 3(2x + 3)$ at $x = 2$

   ANSWER Both give 21

7. Why can $3a$ and $3a^2$ not be collected?

   ANSWER Their letter parts differ

07 Homework

Homework — 5 tasks

Substitute one number into both forms at the end; if they disagree, the simplification is wrong.

1. Simplify $7x + 2y - 3x + 5y$.
2. Expand $4(x + 5)$ and $6(2a - 3)$.
3. Simplify $3(x + 2) + 2x$.
4. Factorise $8x + 12$ and $x^2 + 7x$.
5. Check your answer to task 4 by substituting $x = 3$ into both forms.